MicroStitch — Final MFC Integration Steps

OpenCV 3.4.11 C++ SDK

Use the **latest** `MicroStitchSDK_Handoff.zip` only. Do **not** use the old `Stitcher_ProcessFrame()` / `Stitcher_GetMosaic()` API.

1. Copy SDK files

From the ZIP:

```
MicroStitchSDK_Handoff/
|-- include/
|-- lib/
|   `-- MicroStitch.lib
`-- bin/
    `-- MicroStitch.dll
```

Put them somewhere accessible to the MFC project.

2. Configure the MFC project

The MFC application should be: **Platform:** x64, **Configuration:** Release, **OpenCV:** 3.4.11.

C/C++ → **Additional Include Directories**

```
...\MicroStitchSDK_Handoff\include
C:\tools\opencv3411\include
```

Linker → **Additional Library Directories**

```
...\MicroStitchSDK_Handoff\lib
C:\tools\opencv3411\lib
```

Linker → **Input** → **Additional Dependencies** — add:

```
MicroStitch.lib
```

The MicroStitch SDK was built against OpenCV 3.4.11.

3. Put the runtime DLLs beside the MFC .exe

The application needs:

```
MicroStitch.dll
opencv_core3411.dll
opencv_imgproc3411.dll
opencv_features2d3411.dll
opencv_calib3d3411.dll
```

Put them in the same folder as the MFC executable. The MicroStitch DLL's OpenCV dependencies were verified as these four runtime DLLs.

4. Include the header

```
#include <opencv2/opencv.hpp>
#include <microstitch/stitcher.h>
```

5. Create ONE persistent Stitcher object

Do **not** create it every time a frame arrives. For example, as a member of the MFC dialog/window class:

```
microstitch::Stitcher m_stitcher;
```

Default configuration:

```
microstitch::Stitcher m_stitcher(0.05, 0.35, 150.0);
```

6. Process every microscope frame

Where the MFC application receives its `cv::Mat`:

```
void CYourDialog::ProcessCameraFrame(const cv::Mat& frame)
{
    bool accepted = m_stitcher.processFrame(frame);

    if (m_stitcher.hasMosaic())
    {
        cv::Mat mosaic = m_stitcher.getMosaic();

        // Display mosaic in your second MFC window/control
        DisplayMosaic(mosaic);
    }

    // Display original frame in your first window as usual
    DisplayCameraFrame(frame);
}
```

Important: Do NOT use the old code `Stitcher_ProcessFrame(...)` / `Stitcher_GetMosaic(...)`. Those belong to the previous C-style SDK. The final SDK is C++: `m_stitcher.processFrame(frame); m_stitcher.getMosaic();`

7. Save button

Saving is handled by the MFC app, not the SDK:

```
cv::Mat mosaic = m_stitcher.getMosaic();

if (!mosaic.empty())
{
    cv::imwrite("mosaic.png", mosaic);
}
```

8. Reset button

When starting a new slide:

```
m_stitcher.reset();
```

The next frame becomes the new starting frame.

9. Quit

No MicroStitch-specific quit function is required. The normal MFC application handles closing. The `Stitcher` object is automatically destroyed when its owning MFC class is destroyed.

Final flow

```
Microscope Camera
       |
       v
    cv::Mat
       |
       v
m_stitcher.processFrame(frame)
       |
       v
m_stitcher.getMosaic()
       |
       v
   cv::Mat mosaic
       |
       v
    MFC display
```

Buttons:

```
[Save]  -> cv::imwrite()
[Reset] -> m_stitcher.reset()
[Quit]  -> normal MFC application exit
```

The only API you need

```
microstitch::Stitcher m_stitcher;

m_stitcher.processFrame(frame);
m_stitcher.hasMosaic();
m_stitcher.getMosaic();
m_stitcher.reset();
```

That is the **latest final OpenCV 3.4.11 C++ SDK integration**.